SIMATS ENGINEERING
ASSESSMENT TOOL 2: Scenario-Based

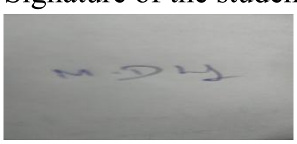
COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	**DHILIBAN M**
Register Number	**192421166**

COURSE OUTCOME COVERED
CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct
I DHILIBAN M/192421166 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student:



Assessment Tasks
Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

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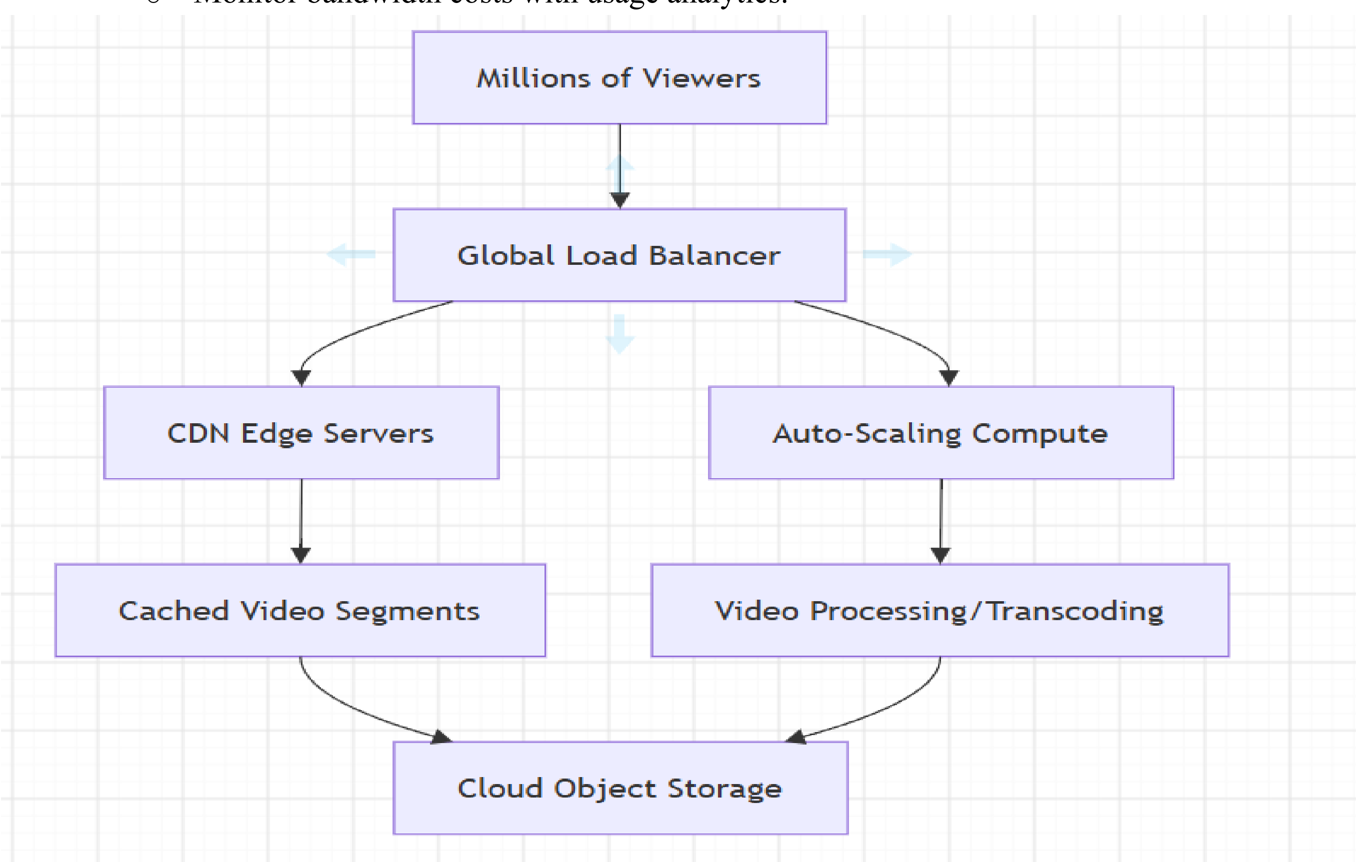
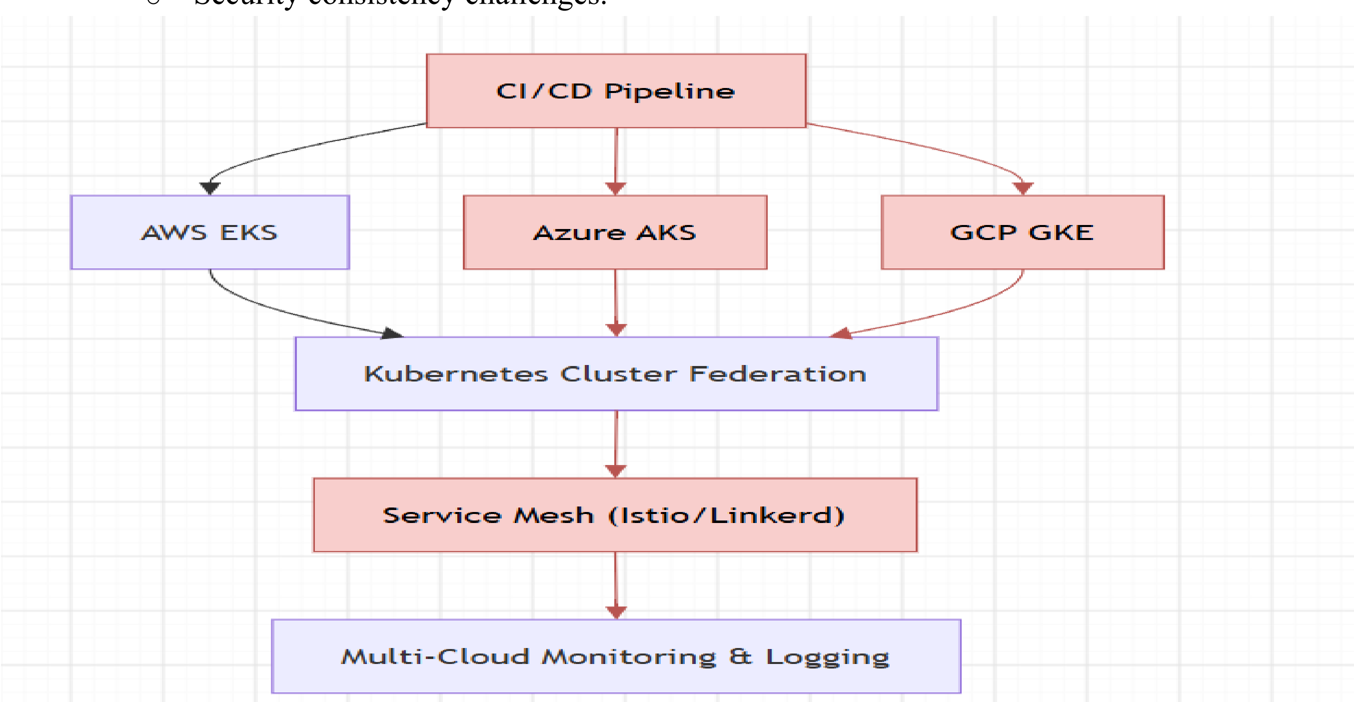
- **Architecture Design:**
    - Use microservices for video ingestion, transcoding, and delivery.
    - Store videos in cloud object storage (AWS S3, GCP Cloud Storage) with lifecycle policies.
    - Integrate CDN (CloudFront, Akamai) for global edge delivery.
    - Employ adaptive bitrate streaming (HLS/DASH) to adjust quality dynamically.
- **Scalability & Caching:**
    - Edge caching reduces latency by serving content closer to users.
    - Use Redis/Memcached for session caching and metadata lookups.
    - Pre-warm caches during peak events (sports, premieres).
- **Load Balancing & Auto-Scaling:**
    - Elastic Load Balancers distribute traffic across compute nodes.
    - Auto-scaling groups adjust capacity based on CPU/network metrics.
    - Multi-region deployment ensures resilience against regional outages.
- **Cost Optimization:**
    - Spot instances for encoding jobs.
    - Tiered storage (hot vs. cold) for popular vs. archived videos.
    - Serverless functions for thumbnail generation.
    - Monitor bandwidth costs with usage analytics.



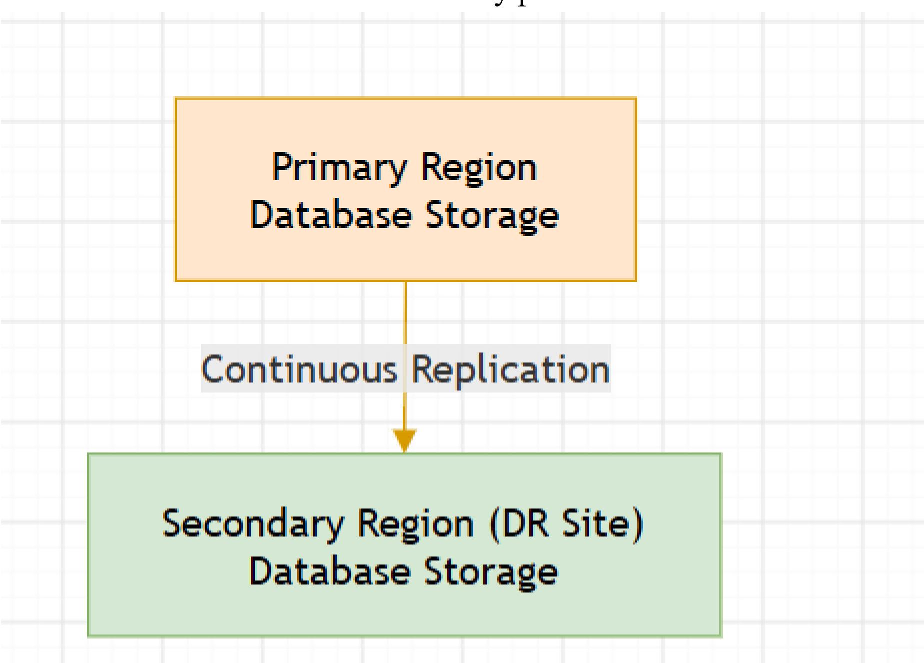
2.A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

- **Solution:**
  - Kubernetes clusters across AWS, Azure, GCP with federation.
  - Helm charts for consistent deployments.
  - CI/CD pipelines (Jenkins, GitHub Actions) for automated builds.
- **Service Discovery & Scaling:**
  - Kubernetes DNS + service mesh (Istio/Linkerd) for routing.
  - Horizontal Pod Autoscaler (HPA) for scaling based on CPU/memory.
  - Cluster Autoscaler for node scaling.
- **Multi-Cloud Strategy:**
  - Use Terraform for IaC across providers.
  - Centralized monitoring/logging (Prometheus + Grafana).
  - Abstract secrets with HashiCorp Vault.
- **Benefits:**
  - Avoid vendor lock-in.
  - High resilience and redundancy.
  - Optimize cost by leveraging cheapest provider for workloads.
- **Risks:**
  - Operational complexity.
  - Inter-cloud latency.
  - Security consistency challenges.



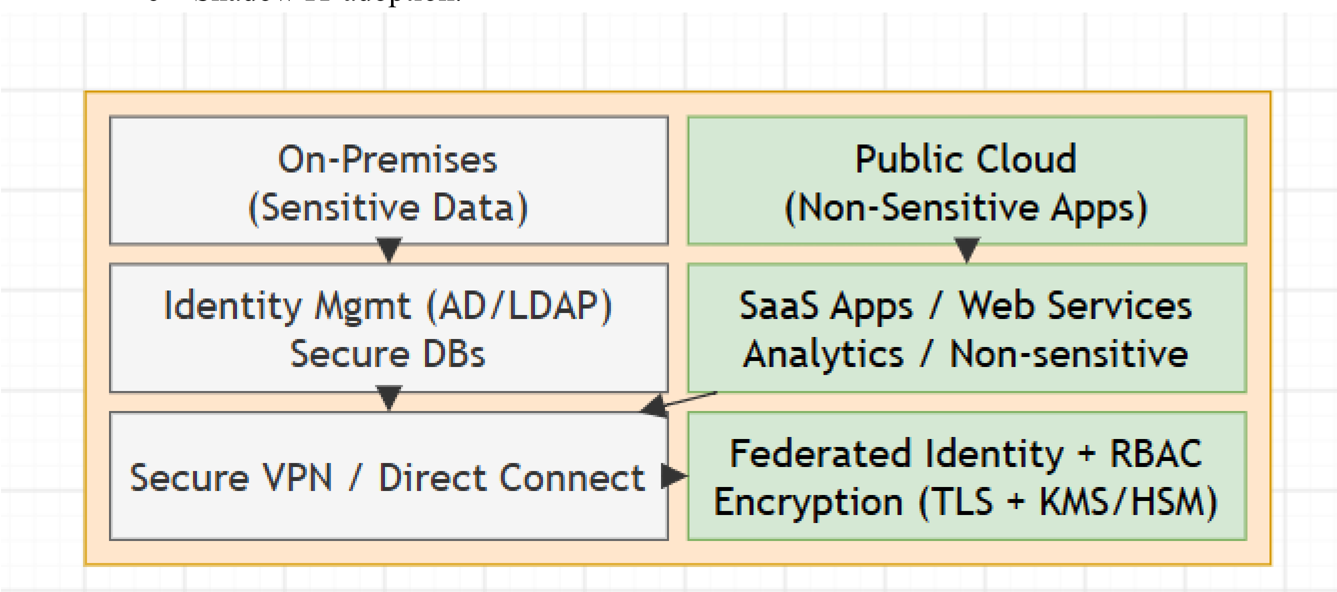
3.An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

- **Solution:**
  - Cloud-native backup services (AWS Backup, Azure Backup).
  - Cross-region replication for critical databases.
  - Object versioning in S3/GCS to prevent accidental deletion.
- **RTO/RPO:**
  - RTO minimized with hot standby and automated failover.
  - RPO achieved via continuous replication and frequent snapshots.
- **Business Continuity Steps:**
  1. Identify mission-critical workloads.
  2. Automate backup schedules with retention policies.
  3. Test recovery drills quarterly.
  4. Maintain secondary region replicas.
  5. Implement monitoring/alerts for backup failures.
- **Risks:**
  - Misconfigured backup policies.
  - Cost of redundant storage.
  - Human error in recovery procedures.



4.A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

- **Architecture:**
  - Sensitive workloads remain on-premises with private cloud.
  - Non-sensitive apps hosted in public cloud.
  - Secure VPN/Direct Connect for communication.
- **Identity Management:**
  - Federated identity (Azure AD, Keycloak) with RBAC.
  - MFA for sensitive access.
- **Encryption:**
  - TLS for data in transit.
  - KMS/HSM for data at rest.
  - Regular key rotation.
- **Challenges:**
  - Policy enforcement across environments.
  - Latency between on-prem and cloud.
  - Hybrid monitoring and patching complexity.
- **Risks:**
  - Misconfigured access controls.
  - Compliance drift.
  - Shadow IT adoption.



5.A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

- **Deployment Strategy:**
  - Active-active multi-region setup with global DNS routing.
  - Traffic managed via Anycast or Route 53 latency-based routing.
- **Data Replication & Failover:**
  - Database replication (Aurora Global Database, Cosmos DB).
  - Automatic failover to secondary region.
  - Synchronous replication for critical data, asynchronous for less critical.
- **Monitoring & Alerting:**
  - Centralized logging (ELK, CloudWatch, Stackdriver).
  - Synthetic monitoring for user experience.
  - Alerts integrated with PagerDuty/Slack.
- **Risks:**
  - Data consistency challenges.
  - Higher operational costs.
  - Complexity in failover testing.